THE PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM ONE TERMINAL EXAMINATION

BIOLOGY

MARKING SCHEME

SECTION A (15 MARKS)

1. **Multiple Choice Questions** (@01 *mark*)

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	C	A	D	B	C	D	B	A	B

2. **Matching Items** (@01 *mark*)

LIST A	i	ii	iii	iv	v
LIST B	A	B	C	D	G

SECTION B (70 MARKS)

3. (a) What is Biology laboratory?

A biology laboratory is a special room or building equipped with scientific instruments, apparatus and chemicals used for conducting biological experiments, investigations and practical activities safely. **(02 $^{1}/_{2}$ *marks*)**

(b) Five qualities of a biology laboratory

1. It should be well ventilated to allow free circulation of air.
2. It should have enough light for clear observation.
3. It should contain clean water supply.
4. It should have safety equipment such as fire extinguisher and first aid kit.
5. It should contain laboratory apparatus and chemicals arranged properly.

(@01 $^{1}/_{2}$ *marks*)

4. (a) Characteristics that distinguish living things from non-living things

Living things:

1. Respire
2. Feed or obtain nutrition
3. Grow and develop
4. Reproduce
5. Excrete wastes
6. Respond to stimuli or sensitivity
7. Move **(@01 *mark*)**

(b) Biology is more than a study of animals and plants. Explain

Biology is broader because it studies all living organisms and their interactions with the environment. It includes microorganisms such as bacteria, fungi and viruses, human beings, heredity, ecology, cells, body functions and relationships among organisms. **(03 *marks*)**

5. Comparison of Terms

(a) Anatomy from Physiology

Anatomy is the study of body structure and organization WHILE Physiology is the study of functions of body parts. **(02 *marks*)**

(b) Cytology from Microbiology

Cytology is the study of cells WHILE Microbiology is the study of microorganisms such as bacteria and viruses. **(02 *marks*)**

(c) Botany from Zoology

Botany is the study of plants WHILE Zoology is the study of animals. **(02 *marks*)**

(d) Unicellular from multicellular organisms

Unicellular organisms consist of one cell only WHILE Multicellular organisms consist of many cells. **(02 *marks*)**

(e) Prokaryotic from eukaryotic cell

Prokaryotic cells are the ones whose nuclear materials are not bound with membrane WHILE Eukaryotic cells are the ones whose nuclear materials are bound with membrane **(02 *marks*)**

6. (a) Procedures of scientific research

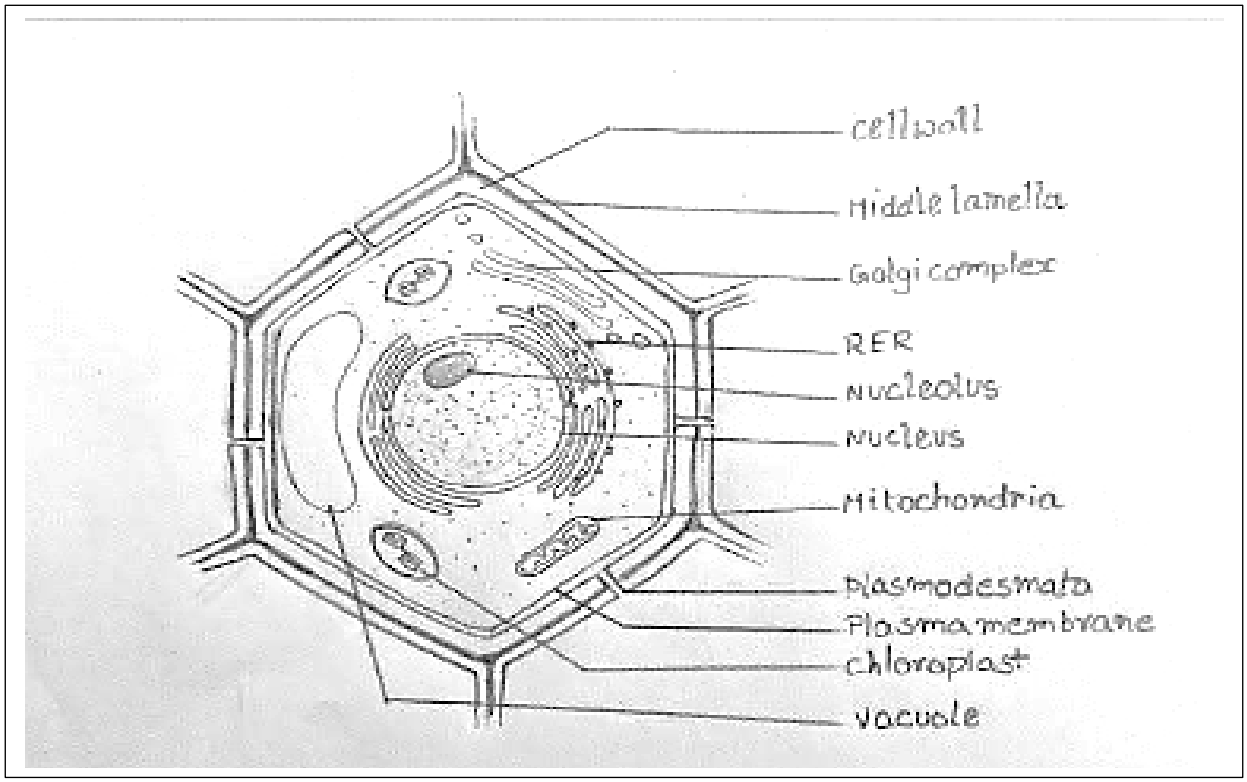
i. Formulation of hypothesis
ii. Observation and data recording
iii. Data analysis and interpretation
iv. Conclusion
v. Reporting results **(@01 $^{1}/_{2}$ *marks* = 07 $^{1}/_{2}$ *marks*)**

(b) If results do not support the hypothesis

If the results do not support the hypothesis, the scientist should reject or modify the hypothesis and conduct another experiment. **(02 $^{1}/_{2}$ *marks*)**

8. (a) Plant cell diagram (labelled)

A DIAGRAM OF PLANT CELL



Title/heading = **01 *mark***

Diagram = **01 $^{1}/_{2}$ *marks***

Labelling = *(any five @* **$^{001}/_{2}$ *marks*** *)*

(b) Differences between animal cell and plant cell

Animal cell	**Plant cell**
Has no cell wall	Has cell wall
Has small vacuoles	Has large vacuole
Has irregular shape	Has fixed regular shape
Lacks chloroplasts	Has chloroplasts
Stores glycogen	Stores starch

(@01 mark =05 marks)

8. (a) Meaning of classification

Classification is the grouping of living organisms according to their similarities and differences.

(02 $^{1}/_{2}$ *marks*)

(b) Reasons why scientists do not prefer artificial classification system

1. It considers only few characteristics.
2. It does not show evolutionary relationships.
3. Unrelated organisms may be grouped together.
4. Related organisms may be placed in different groups.
5. It gives inaccurate scientific information. **(@01 $^{1}/_{2}$ *marks* = 07 $^{1}/_{2}$ *marks*)**

9. (a) Instruments and SI units

i. Mass

- Instrument: Beam balance (**01 *mark***)
- SI Unit: Kilogram (kg) (**01 *mark***)

ii. Time

- Instrument: Stopwatch/clock (**01 *mark***)
- SI Unit: Second (s) (**01 *mark***)

iii. Temperature

- Instrument: Thermometer (**01 *mark***)
- SI Unit: Degree Celsius (°C) or Kelvin (K) (**01 *mark***)

iv. Length

- Instrument: Ruler/metre rule (**01 *mark***)
- SI Unit: Metre (m) (**01 *mark***)

(b) General term for manipulated factor

The general term is variable (independent variable). (**02 *mark***)

SECTION C (15 MARKS)

10. Usefulness of studying Biology

Introduction 01 $^{1}/_{2}$ *marks*

Student should clearly define the term biology, and add some relevant information about it.

Main body (any six points @02 *marks*)

Biology is very important in solving different problems in society in the following ways:

1. It helps in medicine and health care through treatment and prevention of diseases.
2. It improves agriculture by developing better crops and animal breeds.
3. It helps in environmental conservation and protection of natural resources.

4. It helps in food production and preservation.
5. It provides knowledge about proper nutrition and healthy living.
6. Biology helps in biotechnology and genetic engineering.
7. It assists in controlling pests and diseases.
8. It creates employment opportunities such as doctors, teachers, nurses and laboratory technicians.
9. It helps in understanding human body functions.
10. It contributes to industrial production such as vaccines, medicines and fertilizers.

Conclusion. $01\ ^{1}/_{2}\ marks$